

Vol 7 Chapter 2 — Maxwell's Equations from D_IV⁵

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 2 — Maxwell's Equations from D_{IV}^5

Level 1 — one sentence

Maxwell's four equations — Gauss's law for E, Faraday's law, Gauss for B, Ampère-Maxwell — are not independent empirical findings but the substrate's $U(1)$ gauge-field equations in tensor form $\partial_\mu F^{\mu\nu} = J^\nu$ and $\partial_{[\mu} F_{\nu\rho]} = 0$, where $F = dA$ is the gauge curvature 2-form built from the substrate $SO(2)$ connection 1-form A , and the equations follow directly from substrate gauge invariance and the variational principle.

Level 2 — graduate-physicist precision

2.1 The four classical Maxwell equations

Standard form (SI units):

$$\nabla \cdot \vec{E} = \rho/\epsilon_0 \quad (\text{Gauss's law})$$

$$\nabla \cdot \vec{B} = 0 \quad (\text{no magnetic monopoles})$$

$$\nabla \times \vec{E} = -\partial_t \vec{B} \quad (\text{Faraday})$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \partial_t \vec{E} \quad (\text{Ampère-Maxwell})$$

These four equations, plus the Lorentz force law and conservation of charge ($\nabla \cdot \vec{J} + \partial_t \rho = 0$, which follows from Maxwell's equations), describe all classical EM.

2.2 Tensor formulation

The electric and magnetic fields combine into the field strength tensor (a 2-form):

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

The four-current $J^\mu = (c\rho, \vec{J})$.

Maxwell's equations in tensor form:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu \quad (\text{source equations: Gauss + Ampère-Maxwell})$$

$$\partial_{[\mu} F_{\nu\rho]} = 0 \quad (\text{source-free equations: no-monopole + Faraday})$$

Two tensor equations; eight component equations. The second equation (Bianchi identity) is automatically satisfied if $F = dA$ for some 4-vector potential $A^\mu = (\phi/c, \vec{A})$:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

So Maxwell's eight equations reduce to: define F from A ; impose source equation $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$. The Bianchi identity is automatic.

2.3 Maxwell from substrate $U(1)$ gauge action

In BST, the substrate's $U(1)$ gauge sector is the natural $SO(2) \subset K$ factor (Chapter 1). The gauge field A^μ is the substrate's $SO(2)$ connection 1-form. The field strength $F = dA$ is its curvature 2-form.

The Yang-Mills action (Volume 1 Chapter 8) restricted to $U(1)$:

$$S_{EM} = -\frac{1}{4\mu_0} \int d^4x F_{\mu\nu} F^{\mu\nu} = \frac{1}{2} \int d^4x (\epsilon_0 E^2 - B^2/\mu_0)$$

Variational principle: $\delta S_{EM}/\delta A^\nu = 0 \rightarrow \partial_\mu F^{\mu\nu} = 0$ (vacuum) or $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$ (with source).

The Bianchi identity $\partial_{[\mu} F_{\nu\rho]} = 0$ holds automatically because $F = dA$ and $d^2 = 0$.

All of Maxwell's equations are recovered from the substrate $U(1)$ gauge variational principle.

2.4 Why this is a derivation, not a postulate

Standard EM postulates Maxwell's equations as empirical findings. Faraday's law from experiment. Ampère's law from experiment. Maxwell's correction (displacement current) from theoretical consistency. The vacuum speed of light $c = 1/\sqrt{\mu_0\epsilon_0}$ as another empirical fit.

BST derives all four equations from the substrate's natural $U(1)$ gauge structure plus the variational principle on D_{IV}^5 . The constants ϵ_0 and μ_0 have substrate-derivable values (their ratio is c^2 — substrate's natural light-speed); the structure $F = dA$ with the Yang-Mills action $-F^2/4$ is forced by gauge invariance.

This is what "Maxwell from D_{IV}^5 " means: the equations aren't independent inputs; they follow from substrate $U(1) = SO(2) \subset K$ together with the substrate variational principle.

2.5 Gauge invariance

The 4-potential A^μ is not directly observable; only $F = dA$ is. Under a gauge transformation $A^\mu \rightarrow A^\mu + \partial^\mu \Lambda$ for arbitrary scalar function Λ :

$$F_{\mu\nu} \rightarrow \partial_\mu(A_\nu + \partial_\nu \Lambda) - \partial_\nu(A_\mu + \partial_\mu \Lambda) = F_{\mu\nu}$$

(since partial derivatives commute). So F is gauge-invariant. The physical content of EM is in F , not A .

Substrate-mechanism reading: gauge invariance is the substrate's $SO(2)$ symmetry — the substrate's $SO(2)$ phase doesn't enter physical observables; only the $SO(2)$ curvature (gradient) does.

Common gauge choices: - Lorenz gauge: $\partial_\mu A^\mu = 0$ — manifestly covariant - Coulomb gauge: $\nabla \cdot \vec{A} = 0$ — useful for non-relativistic limit - Temporal gauge: $\phi = 0$ — useful for canonical quantization

2.6 Charge conservation

Taking ∂_ν of the source equation $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$:

$$\partial_\nu \partial_\mu F^{\mu\nu} = \mu_0 \partial_\nu J^\nu$$

The left side is zero (antisymmetric F contracted with symmetric $\partial\partial$). So:

$$\partial_\nu J^\nu = 0 \quad \Leftrightarrow \quad \partial_t \rho + \nabla \cdot \vec{J} = 0$$

Charge is conserved. This is automatic from Maxwell's equations — not a separate postulate. Substrate-mechanism: charge conservation is the substrate's $SO(2)$ Noether current (Volume 0 Chapter 8 conservation laws). T2473 (Lyra Friday May 22, 2026) formalizes substrate energy conservation; the EM-charge case is the $U(1)$ analogue.

2.7 Worked example: Gauss's law for a point charge

For a point charge q at origin: integrate Gauss's law over a sphere of radius r :

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0} \implies E(r) \cdot 4\pi r^2 = q/\epsilon_0$$

$$E(r) = \frac{q}{4\pi\epsilon_0 r^2}$$

The Coulomb field. From Maxwell $\leftrightarrow$ Gauss for $\vec{E}$. Standard derivation.

Substrate reading: the point charge is a substrate $SO(2)$ -charged K-type configuration; its $SO(2)$ connection field $A_0(r) = q/(4\pi\epsilon_0 r)$ falls off as $1/r$; the electric field $\vec{E} = -\nabla\phi$ falls off as $1/r^2$ (the substrate's natural Coulomb K-type radial dependence).

2.8 EM waves from Maxwell

Combining Maxwell's source-free equations in vacuum: $\nabla \times \vec{E} = -\partial_t \vec{B}$, $\nabla \times \vec{B} = \mu_0 \epsilon_0 \partial_t \vec{E}$. Taking curl of Faraday:

$$\nabla \times (\nabla \times \vec{E}) = -\partial_t (\nabla \times \vec{B}) = -\mu_0 \epsilon_0 \partial_t^2 \vec{E}$$

Using $\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E}$ (vacuum, $\rho = 0$):

$$\nabla^2 \vec{E} - \frac{1}{c^2} \partial_t^2 \vec{E} = 0$$

with $c = 1/\sqrt{\mu_0 \epsilon_0}$. Wave equation; speed of light c . Same for $\vec{B}$.

This is the foundation of EM waves (Chapter 5).

Substrate reading: EM waves are substrate photon K-type wave packets; the wave-equation derivation falls out of Maxwell, which falls out of substrate $U(1)$ gauge action — all substrate-mechanism.

2.9 K-audit anchors

- T2470 (Lyra Friday May 22, 2026): charge Q from substrate $SO(2)$ weight
- T2473 (Lyra Friday May 22, 2026): energy conservation from substrate Noether (extends to charge for $U(1)$)
- K57 RATIFIED: Bridge Objects (anchor for EM-gauge structure)
- Volume 1 Chapter 8: Yang-Mills gauge action for general G ; $U(1)$ is the abelian case

Level 3 — 5th-grader accessibility

Maxwell’s four equations describe everything classical electricity and magnetism does: - Gauss for E: electric field lines start on positive charges, end on negative - No magnetic monopole: magnetic field lines have no starts or ends; they always loop - Faraday: a changing magnetic field creates an electric field (how generators work) - Ampère-Maxwell: a current OR a changing electric field creates a magnetic field

In BST, these four equations aren’t independent rules. They all follow from one thing: the substrate has a built-in $U(1)$ symmetry (it’s the $SO(2)$ part of the substrate’s structure), and the equations are what you get when you write down “what’s the simplest field theory consistent with that symmetry?” Answer: Maxwell’s. The substrate doesn’t choose Maxwell — it *forces* Maxwell. As a bonus, charge conservation drops out automatically and the speed of light $c = 1/\sqrt{\mu_0\epsilon_0}$ becomes the substrate’s natural EM propagation speed.

What comes next

Chapter 3 develops electrostatics — Coulomb’s law, potential, Gauss’s law applications.

Where to look this up

- Standard Maxwell: Jackson, Ch 1, 6
- Differential-form treatment: Misner-Thorne-Wheeler, *Gravitation*, Ch 4
- Gauge theory: Peskin and Schroeder Ch 15; Quigg, *Gauge Theories*
- BST anchors: T2470 charge, T2473 conservation, K57 Bridge Objects
- Volume 0 Chapter 8: substrate Noether currents
- Volume 1 Chapter 8: Yang-Mills gauge action